

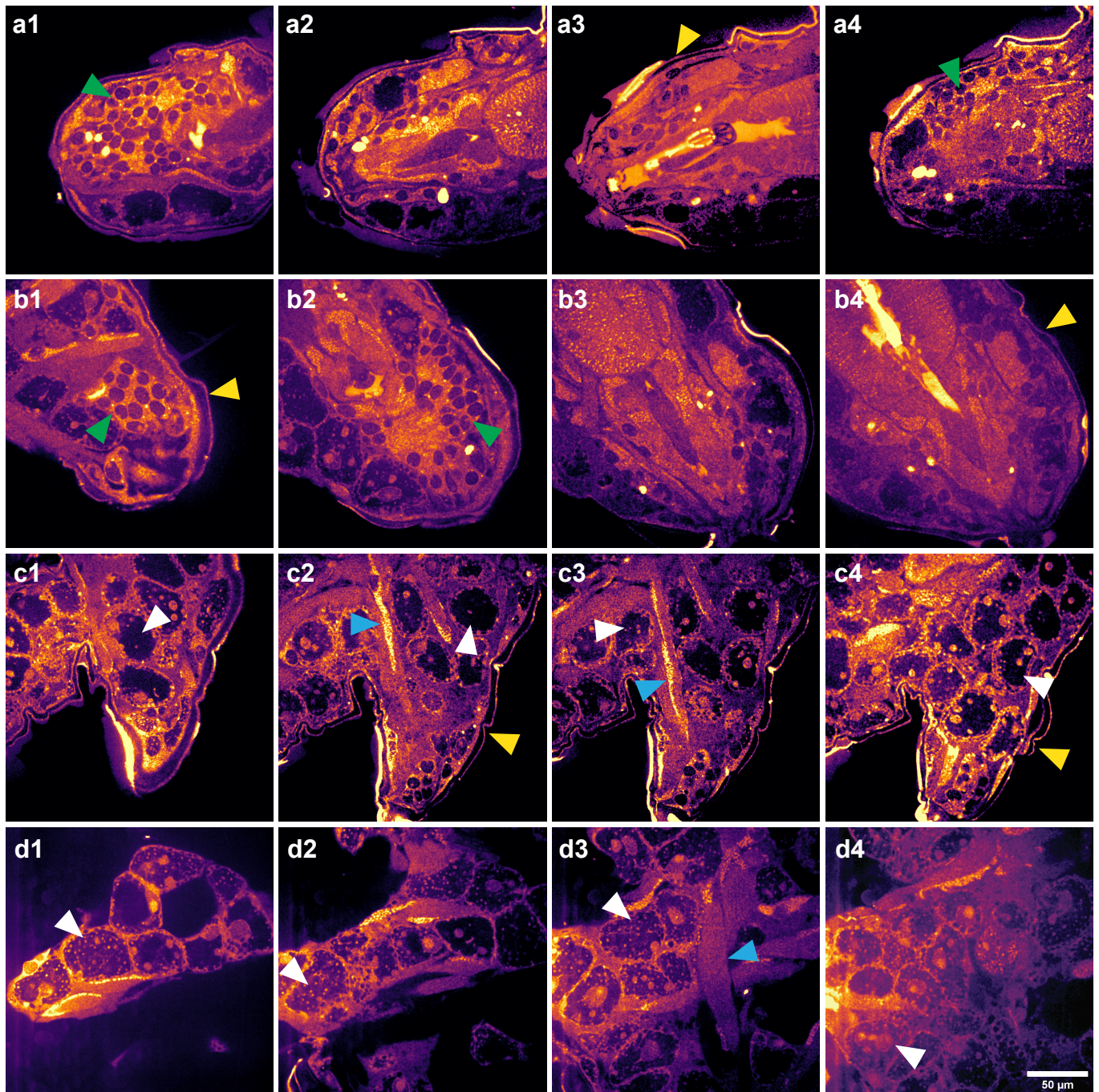


Figure 2. Expansion microscopy of wild-caught tardigrades with total protein labeling.

Wild-caught tardigrades of unidentified species were cryo-expanded and labeled with Alexa Fluor 488 NHS ester, which reacts covalently with primary amines (lysine residues and free N-termini), providing a near-uniform fluorescent signal proportional to local protein density and thus serving as a total protein stain for all tissue compartments. Specimens were imaged by Yokogawa spinning disk confocal microscopy with a CFI Apochromat 60x W NIR, NA 1.0, WD 2.8mm w following approximately 4-fold expansion. Instead of a coverslip, the sample was placed on FEP film (1). (a1–a4) Four sequential z-planes through the head region of tardigrade 1. (b1–b4) Four sequential z-planes through the head region of tardigrade 2. (c1–c4) Four sequential z-planes through a leg of tardigrade 1. (d1–d4) Four sequential z-planes through a leg of tardigrade 2. Across these specimens, multiple anatomical features are distinguishable, including instances of neurons (green arrow), muscle cells (cyan arrow), storage cells (white arrow), and the cuticle (yellow arrow). Arrows indicate examples of these features, not the total present in a given image. Scale bars represent physical distances in the expanded sample and have not been corrected for the expansion factor (~4x).